

LONG YOUTUBE LESSON

From Precise Numbers to Defensible Decisions

Mathematical thinking and reproducible computation before statistics and AI become decisions

Production control	Value
Production unit	PU-B01-C01
Series alignment	SRAI Book 1 — Mathematical Foundations, Chapter 1
Presenter	Mbaye Kebe
Primary audience	Analysts, statistical managers, policy leaders and AI practitioners
Target duration	Approximately 16–18 minutes
Format	Presenter + slides + live notebook demonstration
Version / status	v1.0 — Draft for owner review

0:00	3:30	6:15	10:45
Problem	Framework	Notebook	Decision controls

Core promise By the end of the lesson, the viewer will know how to move from a real problem to a reproducible computation, distinguish verification from validation, and question a result before it influences a consequential decision.

Recording objectives

- Make the first SRAI production unit understandable without oversimplifying it.
- Demonstrate the canonical notebook M1_N01 rather than merely describing reproducibility.
- Connect mathematical discipline to institutional accountability and public decisions.
- Invite viewers to inspect assumptions, limitations and consequences—not only model accuracy.

Production setup

- Record the presenter in a quiet, well-lit setting with the camera at eye level.
- Use 16:9 slides with large text and no more than one central idea per screen.
- Capture the notebook at 1080p; enlarge the code and output so assertions can be read.
- Speak deliberately. Pause after the theory-to-decision chain and after the verification/validation distinction.
- Do not add a repository URL until the public destination has been tested.

0:00–0:50 | Opening question

ON SCREEN / EDITOR Presenter on camera. Title appears after the first sentence: “From Precise Numbers to Defensible Decisions.”

NARRATION

Suppose an analytical team presents an agricultural production forecast showing an eight percent increase. Before the meeting has ended, people are already discussing storage, imports, market measures and public communication.

Then a second analyst discovers that an input file changed, several districts were excluded without explanation, cultivated-area units were inconsistent, and the random seed was never recorded. The figure is still eight percent. But can we still defend the decision built on it?

That is the question behind this lesson: when does a calculated number become trustworthy enough to support a real decision?

0:50–2:00 | Why I begin SRAI here

ON SCREEN / EDITOR Presenter remains on camera. Display a restrained lower-third: “Mbaye Kebe | Statistics, Reasoning and Artificial Intelligence.”

NARRATION

Welcome. I am Mbaye Kebe, and this is the first lesson in SRAI Book 1, Mathematical Foundations.

Over more than three decades working with official statistics, agricultural censuses, large surveys, information systems and decision-support projects, I have seen analytical work fail for reasons that were not mathematically sophisticated. A source file was replaced. A unit changed. An assumption remained in someone’s memory. A model was extended to a population it had never represented.

This is why I do not begin SRAI with the newest algorithm. I begin with mathematical thinking and reproducible computation. Before asking what a system can calculate, we need to be clear about the

reality we are representing, what we have left out, and what decision the calculation is meant to support.

2:00–3:30 | The practical problem

ON SCREEN / EDITOR Show a simple sequence of three screens: “Fast result” → “Hidden assumptions” → “Consequential decision.”

NARRATION

Modern institutions can produce forecasts, risk scores, classifications and dashboards very quickly. Artificial intelligence can also generate a polished explanation in seconds. Speed is useful, but it creates a management problem: the result may travel through the institution faster than anyone verifies whether it is sound.

A result is not defensible merely because the code ran, the graph looks professional, or the model has a familiar name. We need the data source and version, the transformations, variables, assumptions, objective, constraints, implementation, tests, uncertainty and limitations. We also need to know who is accountable for deciding whether the result may be used.

My rule is simple: a result that cannot be reconstructed should not be allowed to carry a major budget, policy or public-communication decision.

3:30–6:15 | The SRAI theory-to-decision chain

ON SCREEN / EDITOR Build the chain one stage at a time. Keep all eight stages visible at the end.

NARRATION

I use an eight-stage chain to slow the analytical process down at the moments where important mistakes are often introduced.

First is reality: what real system, population or decision are we trying to understand? Second is abstraction: which aspects matter, and which have we deliberately omitted? Third is the model: what variables, relationships, assumptions, objectives and constraints represent the problem?

Fourth is the algorithm: what finite procedure converts the model into a result? Fifth is the implementation in Python or another controlled computational environment. Sixth is verification: did we implement and solve the intended model correctly? Seventh is interpretation: what does the output mean, and what does it not mean? Eighth is the decision, where evidence must be combined with context, uncertainty, costs, institutional values and human responsibility.

Notice that Python sits in the middle of the chain, not at the beginning and not at the end. Code cannot decide what reality should be represented, and it cannot assume responsibility for the final institutional choice.

Each stage has its own control question. When the stages are collapsed, different kinds of evidence are confused. A correct calculation is treated as proof of a good model. A good historical fit is treated as proof that a recommendation is fair. A confident explanation is treated as proof that the underlying data are reliable.

6:15–7:10 | Open the companion notebook

ON SCREEN / EDITOR Screen recording begins. Open `M1_N01_mathematical_thinking.ipynb`. Show the title, learning objectives and seed.

NARRATION

Let us now open the companion notebook. It is intentionally small enough for every step to be inspected.

At the top, we record the book and chapter alignment, the status of the notebook and random seed forty-two. We then state the learning objectives. This matters because reproducibility begins before execution. The purpose and conditions of the experiment should be visible to the next analyst, not reconstructed from guesswork.

7:10–8:20 | Record the environment

ON SCREEN / EDITOR Run the environment and seed cell. Pause briefly on the printed output.

NARRATION

The first code cell imports two utilities: one records environment information, and the other sets the random seed. The seed controls pseudo-random operations so that repeated runs begin from the same computational condition.

A seed is not magic, and it does not make every system deterministic. Hardware, library versions, parallel processing and external services can still matter. But recording the seed and environment removes two common sources of unexplained variation and gives another analyst a much better starting point.

8:20–9:50 | Build the simple model

ON SCREEN / EDITOR Show the equation, then the `LinearDemandModel`` class and prediction output.

NARRATION

The worked example defines electricity demand as a linear function of temperature. Demand at temperature T equals a baseline plus a temperature-sensitivity parameter multiplied by T .

The model uses a baseline of one hundred and twenty and a sensitivity of two point five. For temperatures of twenty, twenty-five and thirty degrees Celsius, it predicts one hundred and seventy, one hundred and eighty-two point five, and one hundred and ninety-five units.

This is a model, not reality. The simplicity is deliberate. It allows us to see the variables, parameters, transformation and expected outputs without hiding the reasoning inside a complicated algorithm.

9:50–10:45 | Verify expected behaviour

ON SCREEN / EDITOR Run the assertion cells. Zoom in on “Verification passed” and the `[2.5]` sensitivity output.

NARRATION

Next, the notebook does not merely display the predictions. It compares them with values calculated in advance. The assertion passes only if the computed values agree within a very small numerical tolerance.

The notebook then checks sensitivity. Increasing temperature from thirty to thirty-one degrees must increase modeled demand by exactly two point five units. That test also passes.

These tests are small, but they express a powerful habit: write down what should happen, then ask the computer to test whether it actually happened.

10:45–12:40 | Verification is not validation

ON SCREEN / EDITOR Return to presenter plus a two-column slide: Verification / Validation.

NARRATION

We have now verified important parts of the implementation. But we have not validated the model for real electricity planning.

Verification asks: did we solve the stated model correctly? Validation asks: is the stated model an adequate representation of the relevant reality for the intended use?

The distinction is not academic. Our notebook omits humidity, seasonality, weekdays, economic activity, prices, outages and nonlinear thresholds. Correct code cannot compensate for an inadequate representation of reality.

To validate an operational model, we would need relevant observed data, external comparisons, sensitivity analysis, testing across time and geography, examination of subgroups and rare events, and professional knowledge of the system. We would also need to study the consequences of error. A false alarm and a missed capacity event do not necessarily carry the same cost.

A computation can therefore be reproducible and verified while still being scientifically, ethically or operationally unsuitable. Reproducibility makes the claim inspectable. It does not automatically make the claim true or the decision wise.

12:40–14:25 | From notebook to decision control

ON SCREEN / EDITOR Show seven checklist items one at a time. Use check marks only as questions are answered, not as decoration.

NARRATION

Before relying on an analytical result, I would ask seven questions.

Can another analyst reproduce the result from controlled inputs? Has mathematical and computational correctness been verified? Has the model been validated for the intended reality and population? Are the assumptions, exclusions, uncertainty and limitations visible?

Does the objective reflect the institution's real purpose rather than merely an easy technical metric? Have we examined who experiences the largest error or risk? And finally, is a named human authorized and accountable for the decision?

These questions apply to a conventional statistical model, a machine-learning system, or an answer generated by artificial intelligence. AI may assist at every stage, but it cannot remove responsibility from any stage. It cannot turn an undocumented calculation into knowledge.

14:25–15:45 | Practical exercise for the viewer

ON SCREEN / EDITOR Show the Level A–C exercises and capstone. Highlight “record assumptions and limitations.”

NARRATION

You can now use the notebook at four levels.

At Level A, identify the variable, parameters, assumption and output. At Level B, extend demand to include humidity. At Level C, generate noisy synthetic observations and fit a linear regression. For the capstone, create a capacity alert above a selected threshold, then analyse false alarms and missed alerts.

For every extension, record the seed, dependencies, assumptions, expected behaviour, tests and limitations. Do not treat documentation as something added after the analysis. It is part of the analytical evidence.

15:45–16:40 | Closing message

ON SCREEN / EDITOR Presenter on camera. End with the central statement on screen, then display the SRAI series title.

NARRATION

When I am shown a model result, I do not want to know only what number appeared on the screen. I want to understand the chain of reasoning and evidence that makes the number fit—or unfit—for the decision before us.

That is the foundation we will build on throughout SRAI: mathematical clarity, reproducible computation, honest limitations and accountable human judgment.

If this approach is useful to your work, continue with the companion notebook and share in the comments one analytical result in your organisation that would benefit from a stronger reproducibility record. Thank you for watching.

Post-production package

Recommended primary title

Can You Defend the Number? Mathematical Thinking Before AI Decisions

Alternative titles

- From Precise Numbers to Defensible Decisions
- Reproducible Does Not Mean Valid: A Lesson for Statistics and AI
- Before You Trust an AI Result, Follow This Eight-Step Chain

Thumbnail text

PRIMARY CAN YOU DEFEND THE NUMBER?

Keep the thumbnail to the presenter's face, one number or simple model graphic, and the six-word message above. Avoid equations or small code text.

YouTube chapter markers

Adjust these timestamps against the final edited export before publication.

Time	Chapter
00:00	Can we defend the number?
00:50	Why SRAI begins with mathematical thinking
02:00	The practical institutional problem
03:30	The theory-to-decision chain
06:15	Companion notebook demonstration
07:10	Environment and random seed
08:20	The electricity-demand model
09:50	Numerical verification
10:45	Verification versus validation
12:40	Seven decision-control questions
14:25	Exercises and capstone
15:45	Closing message

Video description

A result can be reproducible and correctly calculated while still being unsuitable for a policy or operational decision. In this SRAI lesson, Mbaye Kebe introduces an eight-stage theory-to-decision chain and demonstrates the canonical M1_N01 notebook using a simple electricity-demand model.

The lesson explains how to record a computational environment, set a random seed, verify expected outputs, test sensitivity, state model limitations and distinguish verification from validation. It concludes with seven questions that analysts and decision-makers can use before accepting a statistical or AI-supported result.

Companion materials: [ADD TESTED PUBLIC LINK AFTER RELEASE]

SRAI: Statistics, Reasoning and Artificial Intelligence

Production Unit: PU-B01-C01

Keywords and hashtags

Keywords: mathematical thinking, reproducible computation, model verification, model validation, statistics, artificial intelligence, decision intelligence, responsible AI, official statistics, Python notebook

#Statistics #ArtificialIntelligence #Reproducibility #DecisionIntelligence #ResponsibleAI #SRAI

Final publication checks

- Run the final notebook from a clean environment and capture the successful execution.
- Confirm that all on-screen output matches the canonical notebook.
- Check captions manually, especially “verification,” “validation,” “reproducibility” and numerical values.
- Replace the description placeholder only with a tested public URL.
- Adjust chapter timestamps after the final edit.
- Confirm music, images, fonts and stock assets are licensed for YouTube use.
- Record owner approval and publication date in SRAI Studio.

Source and control note

Developed from SRAI Book 1, Chapter 1, “Mathematical Thinking and Reproducible Computation,” canonical notebook M1_N01, Executive Brief v1.1 and the PU-B01-C01 GitHub landing page and LinkedIn assets. This v1.0 script remains a controlled draft until Mbaye Kebe completes substantive review and approval.